

Quick Check

Lesson 8: Synthesis, Real-Life Applications, and DQB Review — Solving the Nakuru Mast at Last

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.4: Trigonometry 1 | Lesson 8

A student stands 40 m from the base of a mast and measures the angle of elevation to the top as 50° . The 40 m side is adjacent to the 50° angle, and the height of the mast is the opposite side. Which ratio should be used to find the height?

A $\sin 50^\circ = \text{height} \div 40$

B $\cos 50^\circ = \text{height} \div 40$

C $\tan 50^\circ = \text{height} \div 40$

D Pythagoras' theorem only, since 40 m is a known side

A mast stands on level ground. From a point 40 m from its base, the angle of elevation to the top is 50° . Given that $\tan 50^\circ \approx 1.1918$, what is the height of the mast to the nearest 0.1 m?

A 47.7 m

B 33.6 m

C 30.6 m

D 25.7 m

In the right-angled triangle for the mast problem, the observer's foot is at the 50° angle. The mast is vertical, the ground distance to the mast is 40 m, and the observer looks along a line of sight to the top of the mast. Which statement correctly names the sides relative to the 50° angle?

- A** The 40 m ground distance is the hypotenuse, and the line of sight is adjacent
- B** The 40 m ground distance is adjacent, the mast height is opposite, and the line of sight is the hypotenuse
- C** The 40 m ground distance is opposite, and the mast height is adjacent
- D** The line of sight is opposite, and the mast height is the hypotenuse

A fisherman in a boat on Lake Victoria looks up at the top of a lighthouse and measures the angle of elevation as 30° . A coastguard at the top of the lighthouse looks down at the same boat. Assuming the water level is horizontal, what is the angle of depression from the coastguard to the boat?

A 60°

B 150°

C 90°

D 30°

A boat on Lake Victoria is 120 m from the base of a vertical cliff. The angle of elevation of the cliff top from the boat is 38° . Given that $\tan 38^\circ \approx 0.7813$, what is the height of the cliff above the water to the nearest 0.1 m?

A 93.8 m

B 153.6 m

C 73.9 m

D 94.6 m

While solving a two-triangle height problem, a learner subtracts two heights to find a cliff height and gets a negative answer. What is the best next step?

- A** Assume the calculator is faulty and guess the height
- B** Redraw the diagram and check which angle belongs to which height, since a length cannot be negative
- C** Switch to cosine for every part of the problem
- D** Ignore the minus sign and give the answer as a positive number

The unit driving question asks: 'How can we use the angles and one side of a right-angled triangle to find any unknown height or distance?' Which answer is best?

- A** Use only the tangent ratio in every problem, whatever sides are known
- B** Measure the height directly by climbing with a ladder
- C** Use Pythagoras' theorem, because it links angles to sides
- D** Choose sine, cosine or tangent depending on which sides are known and unknown, then link the known angle and known side to the unknown side

In their final model, a learner writes the complementary-angle relationship $\sin 35^\circ = \cos \theta$. What is the value of θ ?

A 55°

B 35°

C 145°

D 65°

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).